

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

NMTC at PRIMARY LEVEL - V & VI Standards

Saturday, 12 February, 2022

Time : 2 Hrs.

M.M. : 20

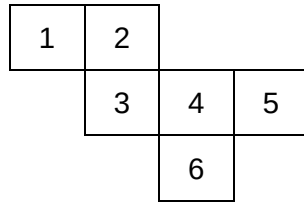
Note:

1. Fill in the response sheet with your Name, Class and the institution through which you appear in the specified places.
2. Diagrams are only visual aids; they are NOT drawn to scale.
3. You are free to do rough work on separate sheets.
4. Duration of the test: **2 hours**.

PART—A

Note

- Only one of the choices A, B, C, D is correct for each question. Shade the alphabet of your choice in the response sheet. If you have any doubt in the method of answering, seek the guidance of the supervisor.
 - For each correct response you get 1 mark. **For each incorrect response you lose $\frac{1}{2}$ mark.**
1. If $(8 * 4) + 6 - (10 \div 5) = 8$, then * stands for the operation.
(a) + (b) $\times$ (c) $\div$ (d) $-$
 2. 20% of a number is equal to 30% of another number. Six times the bigger of these numbers added to the smaller number is 2000. Then 10% of the smallest number is
(a) 10 (b) 20 (c) 30 (d) 25
 3. Samrud chooses a two-digit number. He subtracts it from 200 and doubles the result. The largest number he can get is
(a) 390 (b) 380 (c) 400 (d) 370
 4. The least number which leaves 2 as remainder when divided by 5, 6, 8, 9 and 12 is
(a) 362 (b) 342 (c) 332 (d) 322
 5. Saket takes the number 2021. There are two 2s, one 0 and one 1. Using these numbers, he makes 3-digit numbers, the sum of all these number is
(a) 1488 (b) 150 (c) 1409 (d) 1610
 6. In the net of the adjoining figure numbers 1 to 6 are marked.
The net is folded to form a cube
The number that appears on the opposite face of 6 is



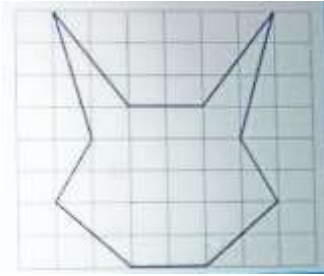
- (a) 4 (b) 3 (c) 2 (d) 1
7. $10^{2021} - 1$ is written as an integer in the usual decimal form. The sum of all digits of this written number is
 (a) 18189 (b) 18187 (c) 18199 (d) 19189
8. August 15th 1992 was a Saturday. What day was 15th August 1991?
 (a) Monday (b) Wednesday (c) Thursday (d) Sunday
9. Two thirds of the students in a class room are seated in three fourths of the chairs. The rest of the students are punished for not submitting the homework and hence asked to stand. There are 6 empty chairs. The number of persons (that is the number of students including the teacher) in the classroom.
 (a) 27 (b) 28 (c) 31 (d) 32
10. There are 24 four digit numbers with different digits, formed by 2,4,5 and 7. One of these four –digit number is a multiple of another. Which one of the following is it?
 (a) 7245 (b) 7542 (c) 7254 (d) 7425

PART - B

Note :

- Write the correct answer in the space provided in the response sheet
 - For each correct response you get 1 mark. For each incorrect response you lose $\frac{1}{4}$ marks.
11. Twelve friends went to a restaurant to have lunch. They ordered for 12 meals. When they were served, they found that the food was too much which can be shared by 18 people. The number of meals they would order to cater to only 12 is _____.
12. The least multiple of 23 which when divided by 18, 21 and 24 leaves the remainders 7, 10 and 13 respectively is _____.
13. The value of $\sqrt{\frac{(0.1)^2 + (0.01)^2 + (0.008)^2}{(0.01)^2 + (0.001)^2 + (0.0008)^2}}$ is _____.
14. Two numbers are respectively 20% and 50% of a third number. The percentage of the first number to the second is _____.
15. Water flows through a rectangular opening 3m × 2m with a speed of 1.5 km/hr. The amount of water that passes through the opening in 5 minutes is $x \text{ m}^3$. Then the value of x is _____.
16. If $A = \frac{1111+3333+5555}{222+333+444}$, $B = \frac{555+222}{1111+2222+4444}$, then $A \times B =$ _____.

17. In the adjoining figure of a cat, each small square is 1 cm^2 . The area of the cat (in cm^2) is



18. The sum of the digits of the number given by the product

$$\frac{666\ldots\ldots 666}{2021 \text{ digits}} \times \frac{999\ldots\ldots 999}{2021 \text{ digits}} \text{ is } \underline{\hspace{2cm}}.$$

19. Two buses X, Y started from two different places P_1 and P_2 respectively, moving towards each other. The ratio of the speed of bus X to that of bus Y was $5 : 4$ after they meet, the speed of bus X was reduced by 20% and the speed of Y was increased by 20%. When bus X arrived at P_2 , bus Y was still 10 km away from P_1 . The distance between the places P_1 and P_2 (in kilometers) is _____.

20. In the adjoining diagram, ABCD is a square P, Q, R, S are the mid points of sides AB, BC, CD and DA respectively. PQ and SR are quadrants of circles with B,D as centres and BP as radius. PS and RQ are part of the circle passing through P,Q, R and S. If $AB = 20 \text{ cm}$, the area of the shaded region (in cm^2) is _____.

